A New Dataset and Recognition for Egocentric Microgesture Designed by Ergonomists

Guangchuan Li[1], Yue Liu[1,2](✉), Weitao Song[1], Cong Wang[1,3], and Yongtian Wang[1]

[1] Beijing Engineering Research Center of Mixed Reality and Advanced Display, School of Optics and Photonics, Beijing Institute of Technology, Beijing, China
liuyue@bit.edu.cn

[2] AICFVE of Beijing Film Academy, Beijing, China

[3] China Academy of Industrial Internet, Beijing, China

Abstract. Virtual and Augmented reality (VR/AR) are widely deployed in industrial, medical, educational, and entertaining fields. The design of interactive interfaces has an impact on usability, comfort, and efficiency. Hand controllers and gestures are popularly used in VR/AR devices. However, users may suffer from overloading on the upper extremities while raising the hand or controller. Therefore, we released a microgesture library with 19 microgestures designed by ergonomists. Users can perform microgestures for an extended duration by resting the forearm on the tables to reduce the load on the upper extremity. Additionally, we collected a microgesture dataset of 2900 samples and utilized the C3D model to recognize the microgesture dataset. Finally, we achieved a recognition accuracy of 93.4% on the microgestures dataset.

Keywords: Microgestures · VR/AR · Gesture recognition · Ergonomist · Dataset

1 Introduction

The acquisition of information for users was different while using VR/AR head-mounted displays (HMDs) compared to traditional devices such as cell phones, tablets, and projectors. The VR/AR techniques have the potential to change the way how we work or entertain in the future [1]. The design of interfaces for HCI is critical to user performance and usability. The interfaces based on hand controllers and mid-air gestures are built to bridge human and VR/AR systems. The use of a hand controller is similar to the operation of the gaming controller so learning difficulty is low for users. Besides, users can signal command with high accuracy, low latency, and tactile feedback while pressing the button on the

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controller. Users may not be able to keep their attention on the hand controller while browsing information with VR/AR devices. Therefore, there are only a few buttons on the controller for users to press to avoid clicking the neighbor button. The interaction efficiency with a hand controller is limited by the number of buttons while users have to browse different menus and shift pages. Additionally, the controller weight is non-trivial and users can not afford to hold it for hours or longer. Moreover, the use of a hand controller contributes to motion sickness so hand gesture [2] as an alternative is attracting the attention of researchers.

Mid-air gestures are deployed in VR/AR devices like HoloLens and Magic Leap et al., and users can perform hand gestures without noticing the hands because of the proprioception. Meanwhile, users can perform dozens of hand gestures to commit commands efficiently without increasing the memory load. Hence, users can perform hand gestures to control the VR/AR devices efficiently. Nevertheless, the existing hand gestures were designed by users without ergonomic background and the repeated formation of hand gestures may be associated with health problems like disorders or even pain in the wrist. Further, users have to raise their hands in front of the camera because hands should appear in the limited field of view of the camera mounted on the devices. The raise of hands will increase the muscle load on the neck and shoulder so users may perceive fatigue on the upper extremities easily. Meanwhile, the lack of feedback has an impact on user input confidence while performing mid-air gestures compared to using a hand controller [3]. Researchers focus on designing microgestures with small hand or finger motions and users can rest the forearm or wrist on the table to reduce the muscle load on the neck and shoulder [4]. Users can interact with smart devices by microgestures for the long term in the future. However, performing microgestures are still potential to contribute to health risk while overextending or over flexing the wrist or finger joints. The microgestures designed by ergonomists can protect users from disorder or pain after performing gestures to interact with smart devices.

The primary purpose of the study was to propose a microgesture set designed by ergonomists for VR/AR systems that may improve usability and comfort. We utilized the deep learning model to recognize the microgesture dataset collected from 9 participants and demonstrated that the application of microgestures can be implemented based on visual methods. The microgestures dataset can be beneficial to researchers who try to design the interfaces based on the microgestures.

2 Realated Work

2.1 The Design of Gestures

The design and selection of gestures for HCIs are crucial to user comfort, and the popular method for designing gestures was to elicit the gestures from users proposed by Wobbrock et al. [5]. The proposed gesture set from users combing user behavior can decrease the cognitive load and improve usability. Rempel et al. [6] found that sign language interpreters have pain or disorders on the upper extremity because their profession is forming mid-air hand gestures repeatedly for many years. Therefore, the formation of gestures from users only may be associated with a health risk. Pereira et al. [7] proposed a 3D hand gesture set for common HCI tasks by evaluating the disorders caused by the hand or finger postures. So, the health risk for users is reduced while performing gestures without awkward postures. However, raising the hands to perform mid-air gestures for a prolonged time tires users on the neck and shoulder easily [8]. Chan et al. [9] proposed a single-hand microgesture set for ubiquitous cameras as accessories integrated on the smart devices everywhere and microgestures were elicited from the non-technical users. The microgestures consist of small finger motion which is hard to be observed by others and users can protect their input privacy while performing microgestures. Nevertheless, users have no experience in ergonomics and part of microgestures can not be performed by users without dexterity fingers. For example, drawing a circle on the palm with a thumb may be a difficult one for users.

2.2 Gesture Recognition

The accuracy and speed for recognizing the hand gestures are key to interaction efficiency and user comfort because high latency and misrecognition may frustrate users and incur wrong responses. The release of commercial depth sensors including Kinect, Leap Motion, and RealSense helps researchers to segment the region of interest like hand region from the clutter background, and the hand gesture recognition and hand pose reconstruction were implemented based on these devices [10,11].

With the increase of dataset samples, deep learning models achieved state-of-the-art performance in visual problems including scene understanding, action recognition, and object segmentation. Therefore, dynamic gestures are utilized to recognize the dynamic gestures. RNN model [12] was used to recognize the dynamic gestures captured with depth, color, and stereo-IR camera from the third view. Wu et al. [13] designed the Dorsum Deformation Network to decode the hand gestures captured by an RGB camera mounted on the wrist to observe the movement of bones, muscle, and tendons of the backhand. Cao et al. [14] developed a recurrent 3D CNN model to recognize dynamic gestures from an egocentric view which demonstrates that the design of interfaces based on egocentric hand gestures captured with an RGB camera could be implemented. Beyond the algorithms, the sample size of the dataset is crucial to the model performance.

The dynamic gesture datasets were released to accelerate the application of hand gestures for HCI systems because the creation of the dataset is time costing and hard for researchers. The egocentric gesture dataset [15] and third view hand gesture dataset [16] was published openly. Moreover, the release of datasets poses a challenge for researchers, and they are inspired to develop algorithms specifically. However, there was a lack of a microgesture dataset. The release of a microgesture dataset can motivate researchers to focus on extracting the features of small hand or finger motion from microgesture sequences which can promote the application of the microgestures for HCI systems.

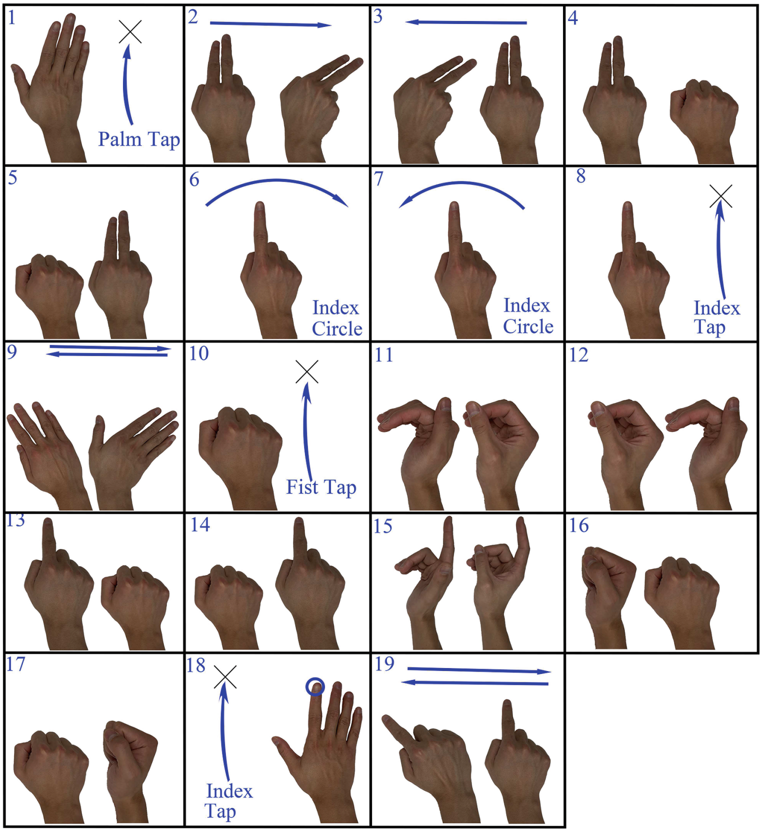

Fig. 1. Microgesture dataset and its literal description. Gesture 1: palm tapping on the table; gesture 2, 3: index and middle fingers swipe right, left; gesture 4, 5: index and middle fingers scratch toward, away; gesture 6,7: index finger moves in circle CW, CCW; gesture 8: index finger tapping on the table; gesture 9: palm scrolling on the table; gesture 10: fist tapping on the table; gesture 11, 12: thumb slides on the side of the index finger to the left/right; gesture 13, 14: index finger scratching toward, away; gesture 15: thumb tapping on the middle finger; gesture 16, 17: fist rotating from the neutral forearm to pronated forearm position or vice-versa; gesture 18: index finger tapping on the table with the palm on the table; gesture 19: index finger scratching on the table.

3 Methodology

3.1 Microgesture Set

The microgestures were designed by ergonomists who have experience in HCIs and health risk management which is different from prior elicitation studies [5,7,9]. The touch or non-touching gestures that appeared in publications or applied in commercial devices were collected in advance for the reason that the existing interaction paradigms impact user preference for gesture selection. Especially, touching devices are widely used in smart devices daily. Additionally, the static gestures were excluded while the systems may respond to them easily while users performing meaningless gestures. All the dynamic microgestures were designed with the forearm in the pronated (palm down) or neutral position (thumb-up) and the overextending or over flexing wrist or finger joints were avoided by following the suggestion from Rempel et al. [6]. Users can rest the forearm on the table while performing the microgestures which can reduce the muscle load on the upper extremities. Hence, users can interact with smart devices by microgestures for an extended duration without being fatigued. Considering the user interaction habits, the different types of microgestures were designed including click, swipe, and circle which are similar to the gestures used for smart devices. For example, users are accustomed to selecting the object by clicking the target on the screen. Therefore, we provided the gesture (*index finger clicks on the supporting surfaces with the forearm in the pronated position*) to complete the selection task. Finally, a microgesture set of 19 microgestuers was proposed and displayed graphically with literation in Fig. 1.

3.2 Microgesture Dataset

Deep learning methods are increasingly used in gesture recognition and the gesture dataset size is vital to the gesture recognition performance and efficiency of designed interfaces. In order to expand the use of proposed microgestures, we created an egocentric microgesture dataset captured by RealSense SR300 which was mounted on the front head to emulate the camera accessorized on the HMDs (Fig. 2). Nine participants were recruited from the university and asked to perform the 19 microgestures repeatedly. Finally, a dataset of 2900 microgestures with RGB and depth information was collected. Gesture 12 (*thumb slides on the index finger from left to right with the forearm in neutral position*) were dis-played in Fig. 3.

3.3 Microgesture Recognition

The proposed microgestures involving small hand or finger motions may be hard to be noticed so we implemented a microgesture recognition based on the deep learning method because of its state-of-the-art performance in action recognition. The 3D convolutional (Conv) layer performs very well in extracting spatiotemporal information from sequential frames which is popular across deep learning models. Tran et al. [17] built a C3D model with eight 3D Conv, five max-pooling, and 2 fully connected layers. The C3D model has more than 17.5 million parameters. All 3D Conv kernels are $3 \times 3 \times 3$ with stride 1 in both spatial and temporal dimensions. Microgetures consist of sequential hand or finger motions; thus we utilized the C3D model to recognize the microgesture dataset. The input shape for the C3D model is $16 \times 112 \times 112 \times 3$ where 16 denotes the frame length, 112 represents the image size, and 3 is the image channel. We set the batch size at 12 and the initial learning rate was set at 0.001. Cross-entropy was used to calculate the loss function while weight regularization was adopted to reduce the overfitting of the model to constrain the increase of the parameters. The image captured by RealSense SR300 with a size of 320 × 140 was reshaped to 128 × 128. To increase the robustness of the model, we cropped a region of 112 × 112 from the reshaped image randomly and the dropout rate was set to 0.5 during the model training. The C3D model was built on TensorFlow and trained with a GPU of 11 GB of memory (GeForce RTX 2080 Ti).

4 Results

80% of microgestures was used to train the C3D model while 20% of microgestures was used to test the C3D model. After training the C3D model to convergence, we achieved a training accuracy of 99.8% and a test accuracy of 88.6%. Therefore, the model was overfitted and the initialization of the model with millions of parameters may have an impact on model performance. The size of the microgesture dataset may limit the learning ability of the C3D model. Hong et al. [18] proposed the transfer learning methods to help researchers to train their model with a small size dataset based on model parameters pre-trained on a larger size dataset. We attempted to use the transfer learning method to improve the C3D model performance on the microgesture dataset. Tran et al. [17] released the C3D model parameters trained on a dataset named with sports 1M dataset of more than one million video clips [19]. We fine-tuned the model parameters with the microgesture dataset based on the parameters published by Tran et al. We achieved a test accuracy of 93.4% with an improvement of 4.8% on the test da-taset. The confusion matrix was displayed in Fig. 4. It was found that 16% of gesture 1 (palm taps on the table) was misrecognized as gesture 10 (fist taps on the table). From the egocentric view, gesture 1 and gesture 10 have the same motion trajectory which contributed to the misrecognition. The recognition accuracy of gestures 2, 3, 11, and 12 was relatively lower because the minor finger motion may be hard to be detected. So, the ability of the C3D model to extract the small difference in actions was not great as to capture large motions. We made an insight to visualize features

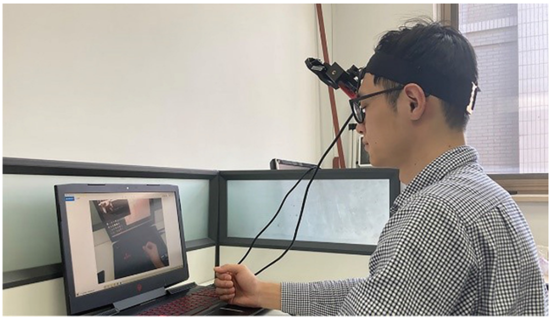

Fig. 2. Microgestures captured with Intel RealSense SR300

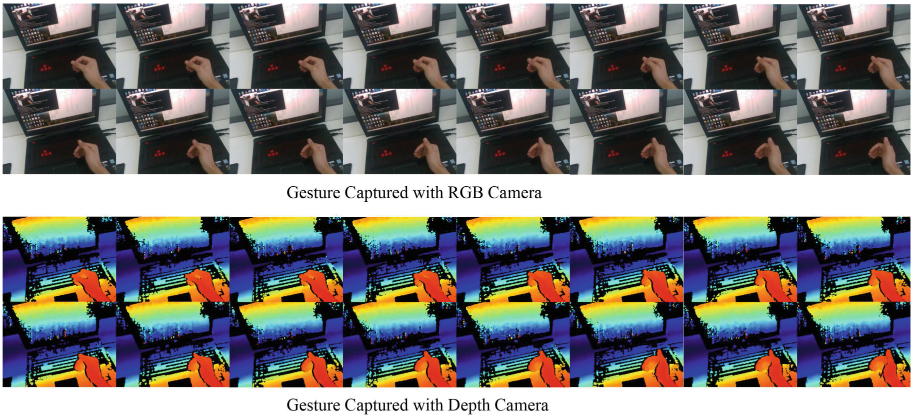

Fig. 3. The gesture 12 displayed in the form of RGB and colored depth

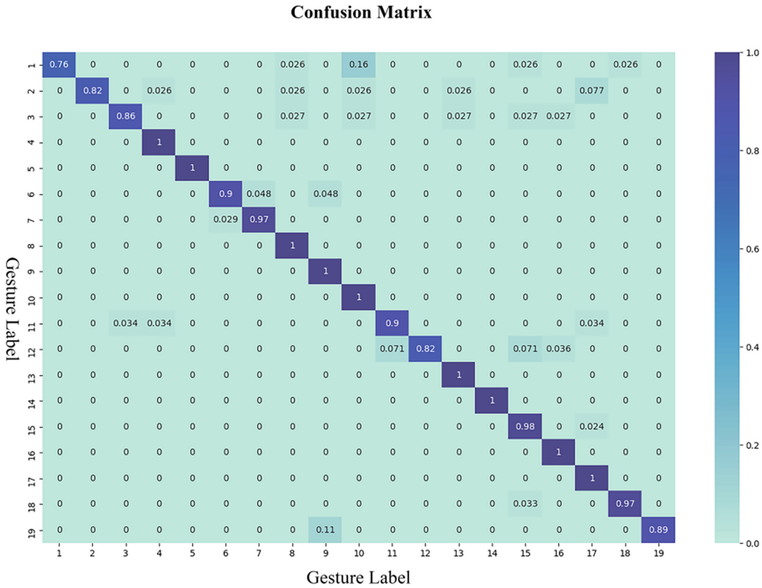

Fig. 4. Confusion matrix for recognizing microgesture training dataset with C3D model.

while the C3D model recognizing the gesture 12 (thumb slides on the index finger with the forearm in neutral position) and showed the features in Fig. 5. On layers 3, 4, and 5, the feature map showed that the direction of the thumb can be captured by the model and the spatial motion of the thumb was extracted as the high-level feature which was visualized in layer 7.

Fig. 5. The visualized feature of the 3D CNN layers from the C3D model while recognizing the gesture 12 (thumb sliding on the index finger from left to right).

5 Conclusion

AR/VR devices are potential to entertain, educate or train users while the interfaces has an impact on usability, comfort and interaction efficiency. To solve the problem that users may suffer from overload on the neck and shoulder while performing mid-air gestures or using a hand controller, A microgesture set designed by ergonomists was presented in this study to reduce the load and health risk to upper extremities while resting the arm on the supporting surface. Therefore, users can perform microgestures to control smart devices for an extended duration without being fatigued or induried. Besides, the minor finger or hand motion is not easy to be observed by others so interaction privacy is protected.

The microgesture dataset is vital to researchers and developers while designing the interfaces based on such microgestures. We created a microgesture dataset of more than 2900 samples captured with RGB and depth cameras. The C3D model was utilized to recognize the proposed microgestures and we achieved a recognition accuracy of 93.4% on the test part. Therefore, the application of microgestures for smart devices can be implemented based on the current deep learning methods visually. The release of the microgesture dataset and the trained C3D models will be beneficial to HCI researchers who may lack knowledge in complicated image processing algorithms and deep learning models. However, we observed that the ability of the C3D model to extract minor finger motions should be improved from the confusion matrix on the recognizing the microgesture dataset (Fig. 4). How to extract the small finger motion from the egocentric view visually is still a problem? So, the release of the microgesture dataset provides a new challenge for researchers to design the algorithms specifically to recognize the finger or hand motions accurately.

References

1. Guo, J., et al.: Mixed reality office system based on Maslow's hierarchy of needs: towards the long-term immersion in virtual environments. In: 18th IEEE International Symposium on Mixed and Augmented Reality, Beijing, pp. 224–235. IEEE (2019)
2. Saredakis, D., Szpak, A., Birckhead, B., Keage, H.A.D., Rizzo, A., Loetscher, T.: Factors associated with virtual reality sickness in head-mounted displays: a systematic review and meta-analysis. Front. Human Neurosci. **14**, 96 (2020)
3. Wang, Y., MacKenzie, C.L.: The role of contextual haptic and visual constraints on object manipulation in virtual environments. In: CHI 2000: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, pp. 532–539. ACM (2000)
4. Visser, B., Korte, E.D., Kraan, I.V.D., Kuijer, P.: The effect of arm and wrist supports on the load of the upper extremity during VDU work. Clin. Biomech. Supp. **1**(15), 34–38 (2000)
5. Wobbrock, J.O., Morris, M.R., Wilson, A.D.: User-defined gestures for surface computing. In: CHI 2009: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, pp. 1083–1092. ACM (2009)
6. Rempel, D., Camilleri, M.J., Lee, D.L.: The design of hand gestures for human-computer interaction: lessons from sign language interpreters. Int. J. Hum.-Comput. Stud. **10**(72), 728–735 (2014)
7. Pereira, A., Wachs, J.P., Park, K., Rempel, D.: A user-developed 3-D hand gesture set for human-computer interaction. Hum. Fact. **57**(4), 607–621 (2015)
8. Hincapié-Ramos J.D., Guo X., Moghadasian P., Irani P.: Consumed endurance: a metric to quantify arm fatigue of mid-air interactions. In: CHI 2014: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, pp. 1063–1072. ACM (2014)
9. Chan, C., Seyed, T., Stuerzlinger, W., Yang, X., Maurer, F.: User elicitation on single-hand microgestures. In: CHI 2016: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, pp. 3403–3414. ACM (2016)
10. Nai, W., Liu, Y., Rempel, D., Wang, Y.: Fast hand posture classification using depth features extracted from random line segments. Hum. Fact. **65**, 1–10 (2016)
11. Mueller, F., et al.: Real-time pose and shape reconstruction of two interacting hands with a single depth camera. ACM Trans. Graph. **38**(4), 1–13 (2019)
12. Molchanov P., Yang X., Gupta S., Kim K., Tyree S., Kautz J.: Online detection and classification of dynamic hand gestures with recurrent 3D convolutional neural networks. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4207–4215. IEEE (2019)
13. Wu W., Yuan Y., Yeo H., Quigley A., Koike H., Kitani K.M.: Back-hand-pose: 3D hand pose estimation for a wrist-worn camera via dorsum deformation network. In: Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology, pp. 1147–1160. ACM (2020)
14. Cao, C., Zhang, Y., Wu, Y., Lu, H., Cheng, J.: Egocentric gesture recognition using recurrent 3d convolutional neural networks with spatiotemporal transformer modules. In: 2017 IEEE International Conference on Computer Vision (ICCV), pp. 3763–3771. IEEE (2017)
15. Zhang, Y., Cao, C., Cheng, J., Lu, H.: EgoGesture: a new dataset and benchmark for egocentric hand gesture recognition. IEEE Trans. Multimedia **5**(20), 1038–1050 (2018)

16. Materzynska J., Berger G., Bax I., Memisevic R.: The jester dataset: a large-scale video dataset of human gestures. In:2019 IEEE/CVF International Conference on Computer Vision Workshop (ICCVW), pp. 2874–2882. IEEE (2019)
17. Tran, D., Bourdev, L., Fergus, R., Torresani, L., Paluri, M.: Learning spatiotemporal features with 3D convolutional networks. In: 2015 IEEE International Conference on Computer Vision (ICCV), pp. 4489–4497. IEEE (2015)
18. Ng, H., Nguyen, V.D., Vonikakis, V., Winkler, S.: Deep learning for emotion recognition on small datasets using transfer learning. In:17th ACM International Conference on Multimodal Interaction, pp. 443—449. ACM (2015)
19. Andrej K., George T., Sanketh S., Thomas L., Rahul S., Li F.: Large-scale video classification with convolutional neural networks. In: 2014 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 1725–1732. IEEE (2014)